

Veil

Private secured lending on Canton — with collateral locked by the Canton Coin token standard itself.



Live on the HackCanton DevNet node · veil-lite-hackathon.vercel.app · no sign-up needed
No Witness Labs

Secured lending needs privacy and enforceable collateral

Public chains leak the book

Counterparties, sizes, collateral and liquidation levels become public. Institutions can't run a repo or credit desk like that.

Off-chain is fragmented

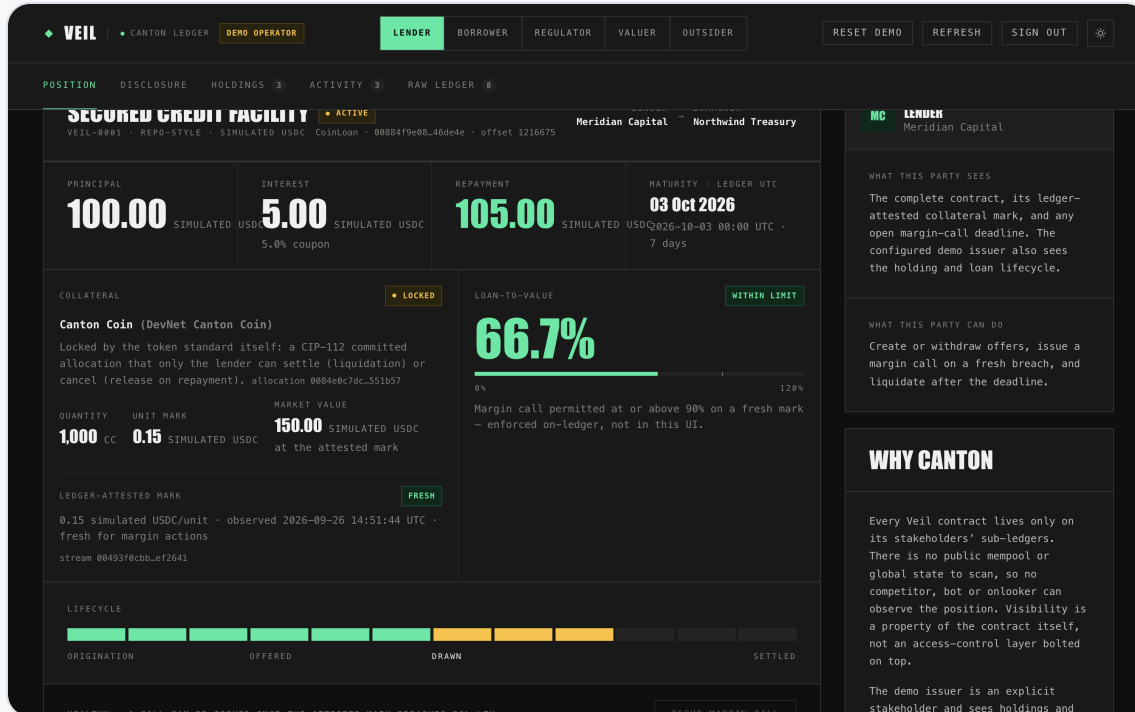
Terms, price checks, margin calls and settlement live in emails, spreadsheets and custodian systems — nobody holds one enforceable state.

Veil gives two known counterparties one private, enforceable loan: the ledger checks every transition, and only the parties who need to see a contract can.

A complete loan lifecycle, enforced on-ledger

Stage	What Canton enforces
Offer	Lender pre-funds the principal and chooses the terms; the borrower can reject it, and it can expire
Accept	Fresh attested price must show LTV below threshold; collateral locks and principal is delivered in one transaction
Margin call	Only on a fresh breach; a ledger-time deadline. Cure by top-up, cash pay-down , or price recovery
During the loan	Partial repayment; collateral substitution (T-Bill → MMF) with lender approval and an LTV re-check
Close	Repay releases collateral; liquidation only after an expired call on a breached mark, or after maturity — and the lender takes only what covers the debt, the surplus returns to the borrower

Real Canton Coin collateral, locked by CIP-112



A committed allocation, not an escrow account

On acceptance the borrower's Canton Coin is locked in a **Token Standard V2 committed allocation** that the lender and the borrower execute jointly, so neither can settle or release it outside Veil.

✓ Borrower's early withdrawal **refused by Canton Coin itself**

✓ Repay → allocation cancelled, coin unlocks

✓ Liquidation → settled to the lender via **SettleBatch**

✓ Verified on hackcanton-01 with real DevNet CC

Privacy and atomicity are structural, not bolted on

NEED-TO-KNOW

Lender, borrower and regulator see the loan; the valuer sees only its prices; an outsider's ledger query returns `[]`.

ATOMIC

Lock collateral + check LTV + deliver principal + open the loan: one multi-party transaction, or nothing.

COMPOSABLE

Collateral uses the network's own token standard (CIP-112) — the same primitive any Canton asset can plug into.

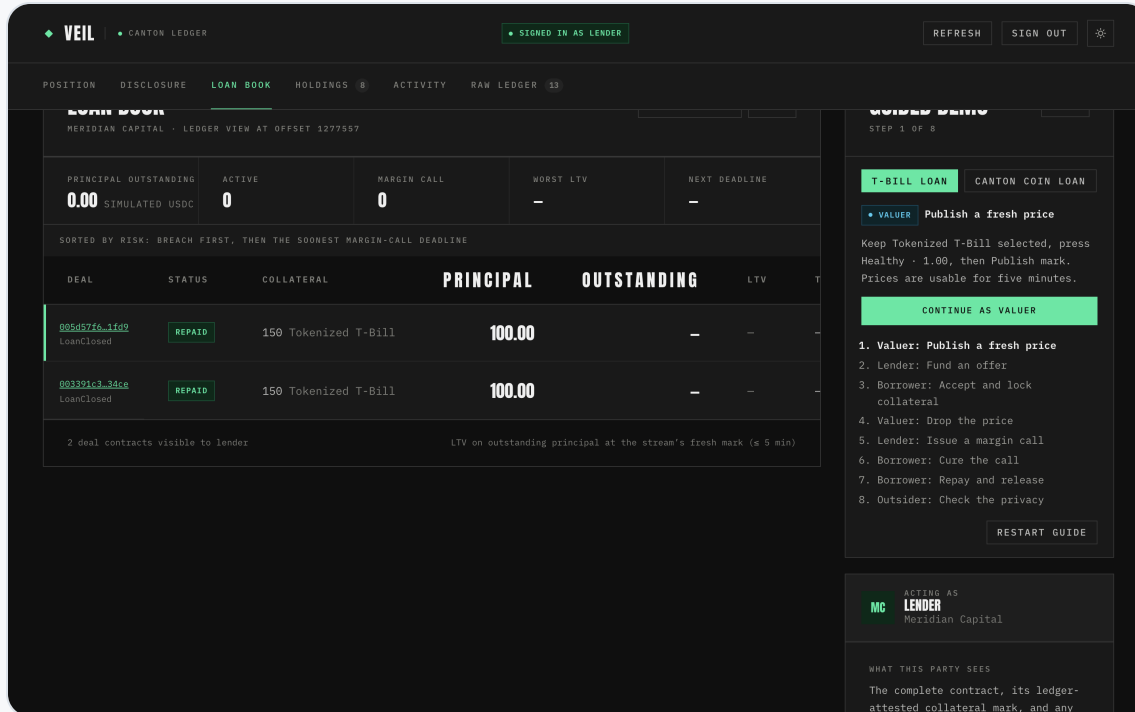
The in-app Disclosure tab shows the signatory/observer matrix for every contract; the Raw ledger tab shows exactly what each party's ledger query returns.

Every audit finding triaged, the valid ones fixed

Finding (auditor severity)	Outcome
Lender-supplied liquidation price (High) · unissued holdings (Medium) · exact repayment (Low)	Already fixed earlier this season
Liquidation seizes surplus collateral (Medium)	Fixed in 0.9.0: only the debt is covered, the rest returns (T-Bill loans)
No offer reject / expiry · unbounded terms · lender-only record deletion · regulator choice	Fixed in 0.9.0 with a regression test each
Canton Coin liquidation takes the whole locked leg	Documented limitation: needs iterated settlement
Second round: lender could settle the coin lock alone; unauthenticated allocation factory (High)	Fixed in 0.10.0: joint executors, verified allocations. Verified live on DevNet

Full triage with evidence: docs/AUDIT-TRIAGE.md. Each release (0.9.0, 0.10.0) was checked as a compatible upgrade of the package already on DevNet before it was uploaded.

Guided demo, loan book and keeper



Guided demo

Walks a first visit through the T-Bill or Canton Coin loan. Progress is read from each party's ledger view; one click switches party.

Loan book

Every visible loan ranked by risk, with a CSV/JSON export of that party's view for the regulator.

Keeper

Timely margin calls and liquidations, including Canton Coin. Journals every command, recovers lost responses from the ledger, optional maker-checker.

Who moves what, and why

Lender earns fixed interest on principal it pre-funds; protected by a locked, enforceable claim on collateral.

Borrower gets liquidity without selling its assets; can cure calls with cash or more collateral, or swap collateral.

Valuer publishes signed prices on a stream both counterparties authorised.

Regulator observes positions and settlements without taking part.

NETWORK ACTIVITY PER LOAN

Each loan is 6–10 ledger transactions: funding, a token-standard allocation, price updates, margin actions, and a token-standard settlement or cancellation.

Not built: protocol fees, valuer fees, stablecoin principal (principal is simulated USDC today).

Small, auditable, deployed

Layer	What it is
Contracts	Daml package <code>veil-lite</code> 0.10.0 on hackcanton-01; Token Standard V2 interfaces (exact vetted packages); optional <code>veil-price-committee</code> for a 2-of-3 decentralized valuer (BitSafe DecMan, LocalNet)
Server	Vercel functions: per-party sessions, per-party read/act checks, read-only token-registry proxy; request IDs, sanitised ledger errors, timeouts, and lost-submission checks against the ledger
App & ops	React UI (role views, guided demo, loan book, disclosure matrix, raw ledger) · Node keeper
Evidence	76 Daml scripts · 144 server, keeper, UI and wallet-verifier tests · DevNet runs with real CC, including a maker-checker liquidation and a recovered lost submission

Trust boundary, stated plainly: on the shared node one ledger user hosts all demo parties, so our server enforces role separation there; on DevNet valuations are manually attested (the decentralized price committee runs on LocalNet); T-Bill/MMF and USDC are simulated.

Two minutes on the live app

Easiest: open the app, pick any party, and follow the Guided demo panel.

CANTON COIN LOAN

1. Valuer → Canton Coin → publish 0.15
2. Lender → collateral **Canton Coin (real)** → Create offer
3. Borrower → Accept (1,000 CC locks)
4. Valuer → 0.11 · Lender → Issue margin call
5. Borrower → Pay down 30 → Repay (CC unlocks)

PRIVACY CHECK

Enter as **Outsider** → Raw ledger: `[]`.

Enter as **Valuer**: prices only, no loan.

Enter as **Regulator**: the loan and its settlement, read-only.

Open access: pick a party and click Enter.

From demo to pilot

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|----|-------------------------------|--|
| 01 | Validate the user | Walk 3 lending / treasury operators through one real deal. No customer evidence yet. |
| 02 | Real signing | A borrower whose key stays with the user now runs on DevNet, verifying each transaction before signing (onboarded by the node operator). Next: bring it into the web app, and connect existing Canton wallets through PartyLayer (CIP-0103). |
| 03 | Real assets and prices | Stablecoin principal (e.g. Brale) and tokenised T-Bills via the token standard; the decentralized price committee (built, on LocalNet) fed by market data such as Kaiko or Coin Metrics. |